

K-Nearest Neighbors

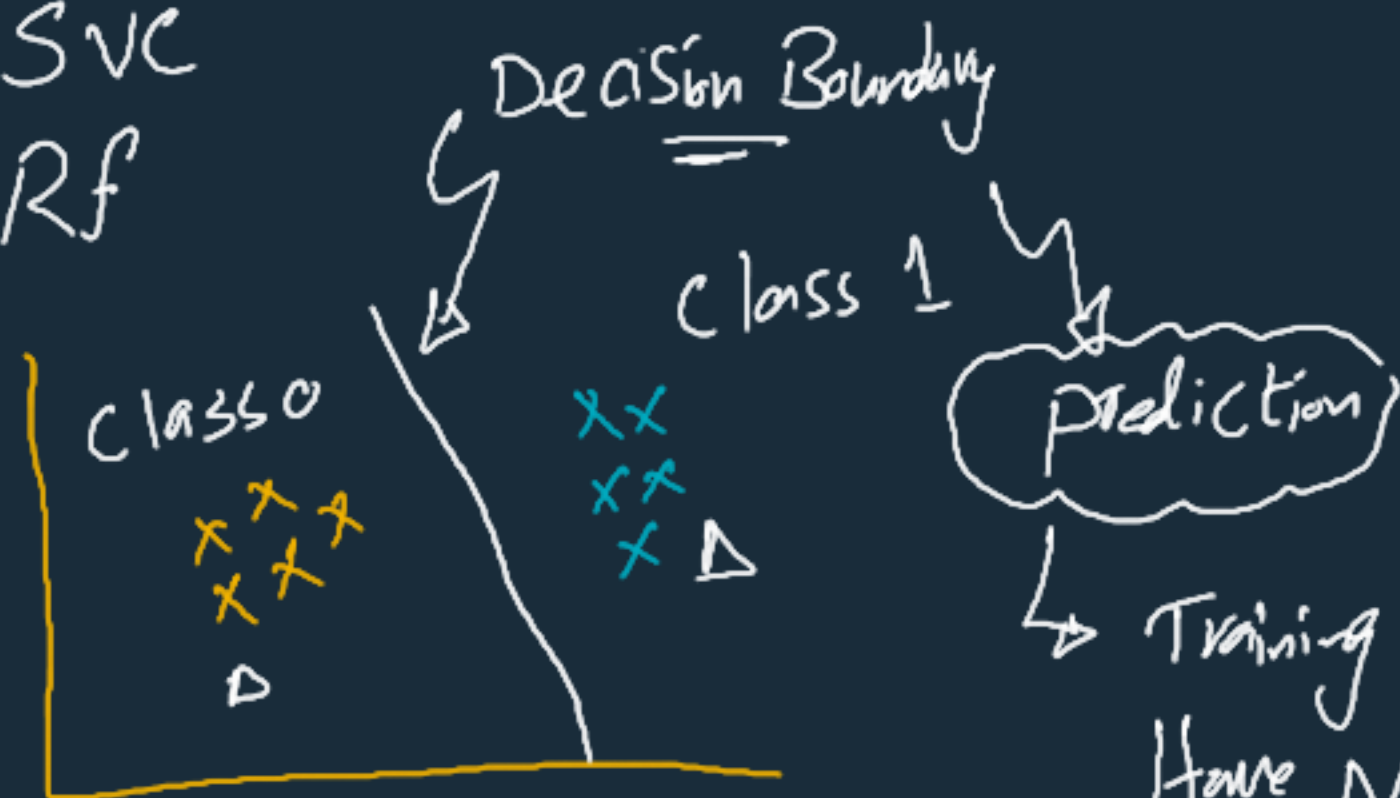
ML Models

Model Based

↳ Logistic Regression

SVC

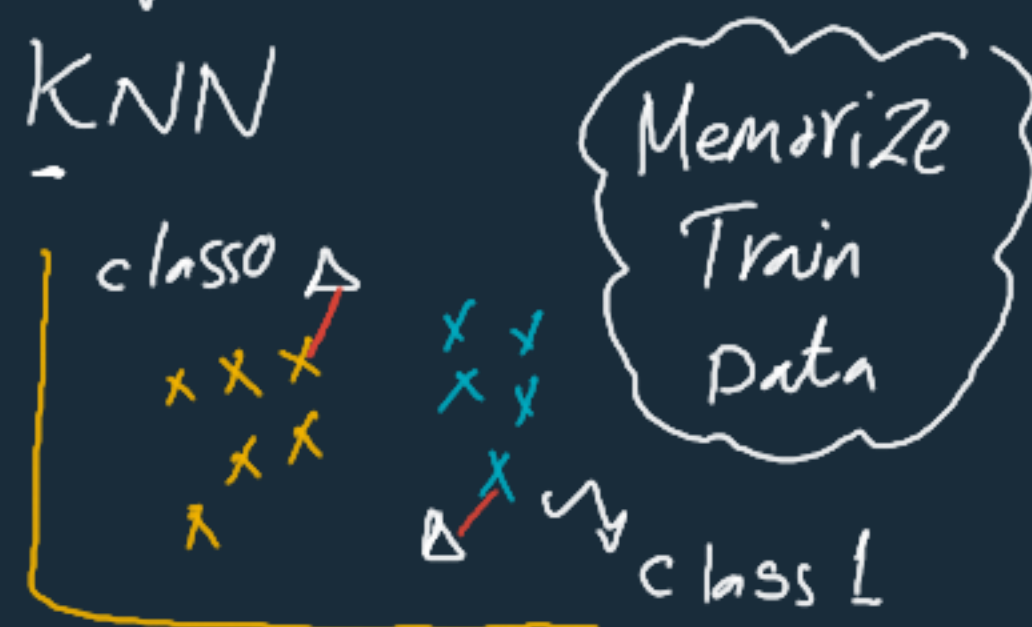
Rf



↳ Training points
Have NO Effect

Instance Based

↳ KNN



prediction depends on
Training points

Distance $\rightarrow$ Measure of similarity



Age $\left\{ \begin{array}{l} \rightarrow \text{class 0 } \geq 5\text{yrs} \\ \rightarrow \text{class 1 } < 5\text{yrs} \end{array} \right.$

① Distance Metrics ?

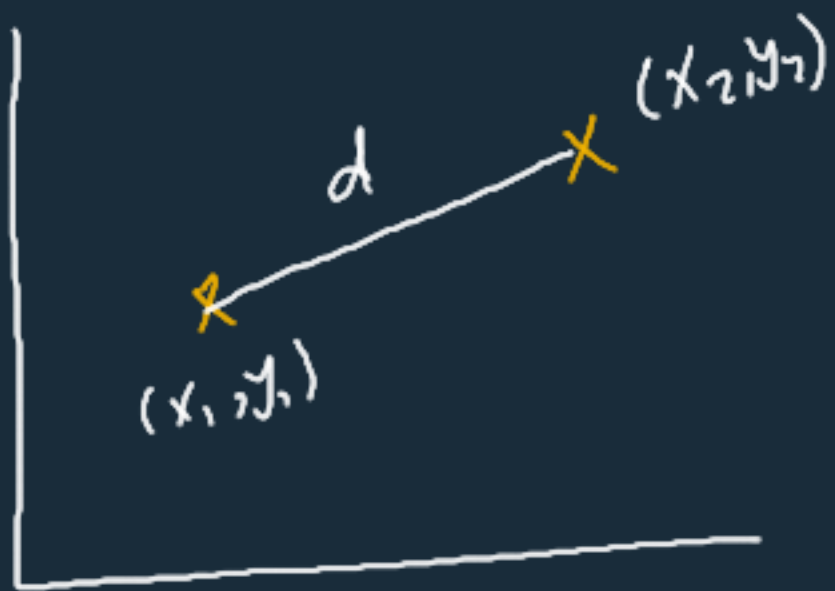
② K-value ?

① Distance Metrics

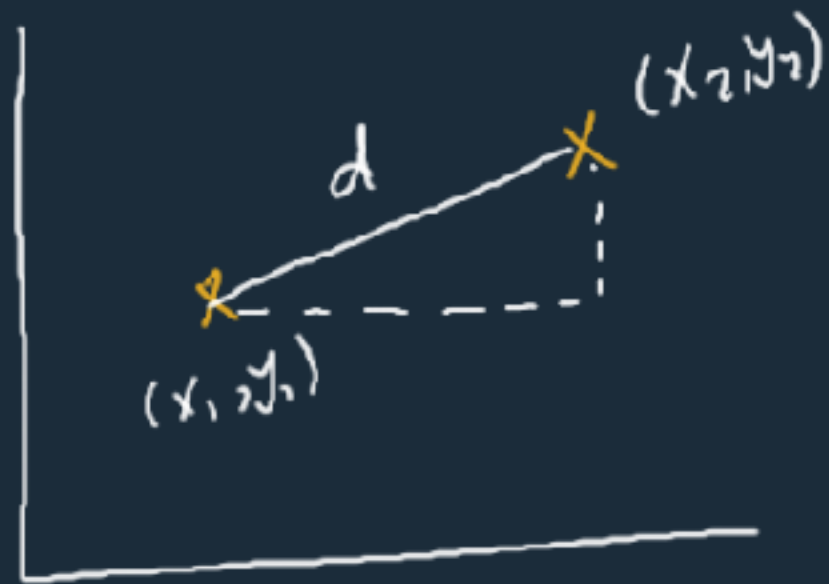
Euclidean

Manhattan

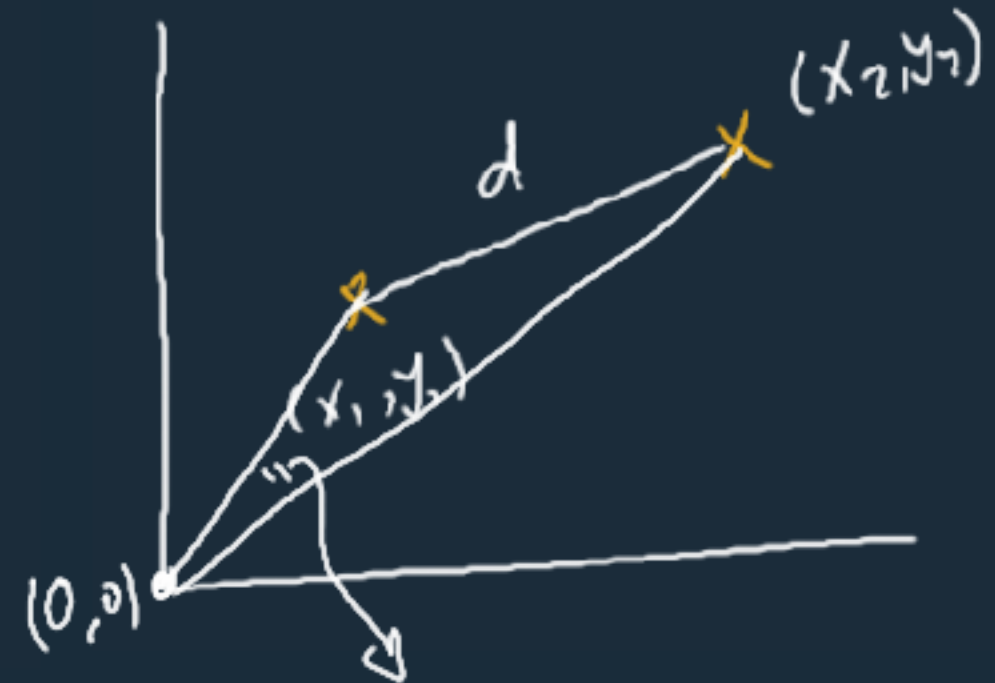
Cosine Similarity



$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



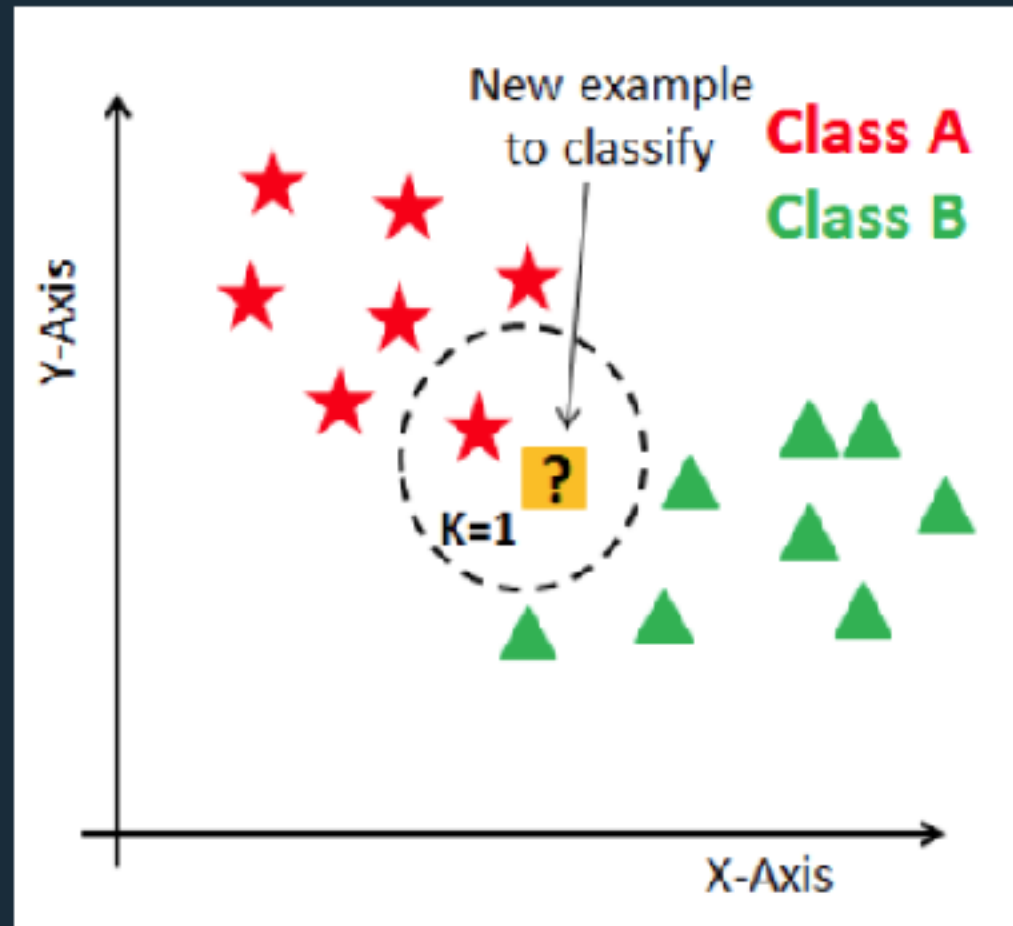
$$d = (x_2 - x_1) + (y_2 - y_1)$$



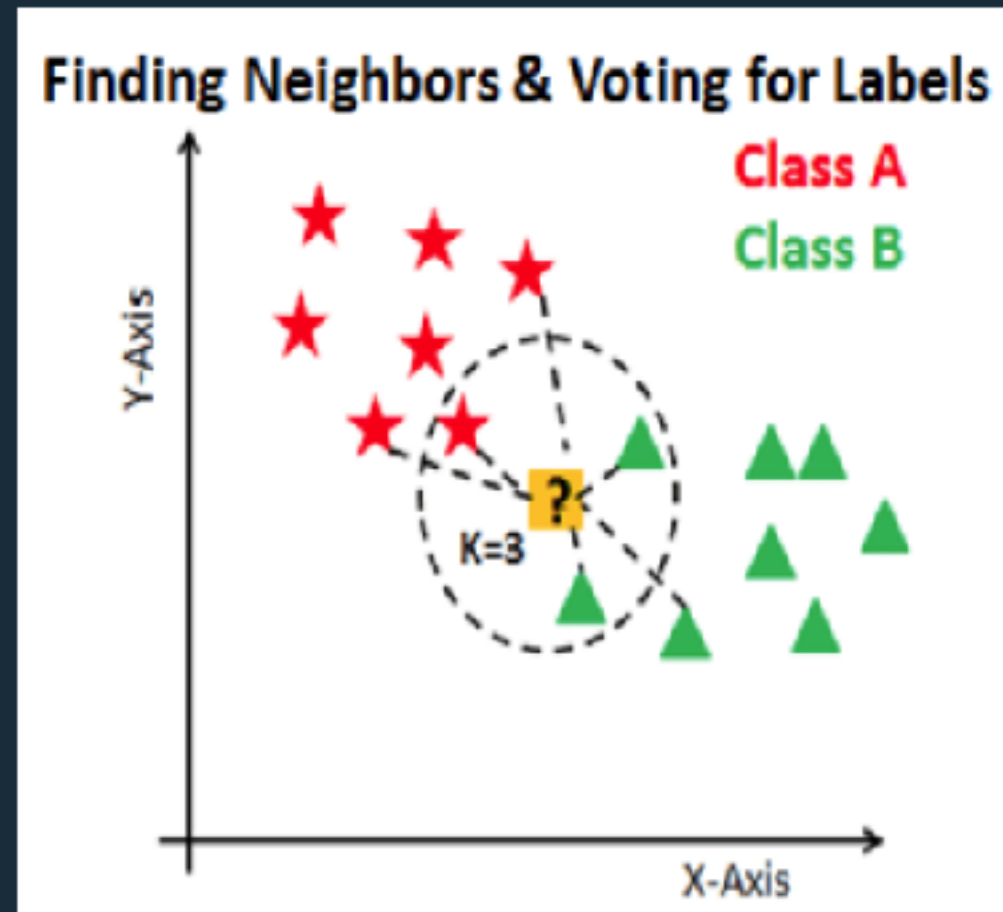
$$\cos \theta$$

② K-Value

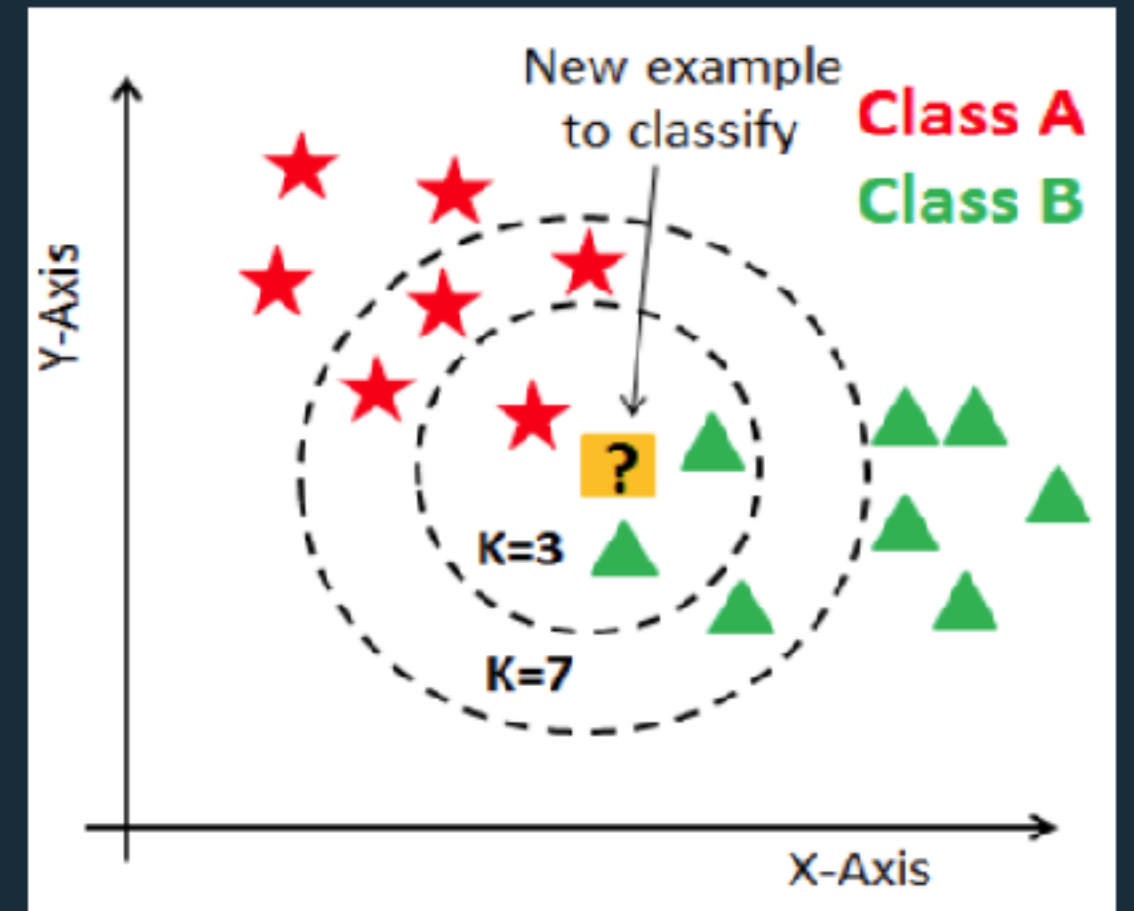
K=N (underfitting)



$K=1$
(overfitting)



Voting
 $\left\{ \begin{array}{l} 2 \rightarrow B \\ 1 \rightarrow A \end{array} \right\} \textcircled{B}$



Voting
 $\left\{ \begin{array}{l} 4 \rightarrow A \\ 3 \rightarrow B \end{array} \right\} \textcircled{A}$

Knn

Classifier

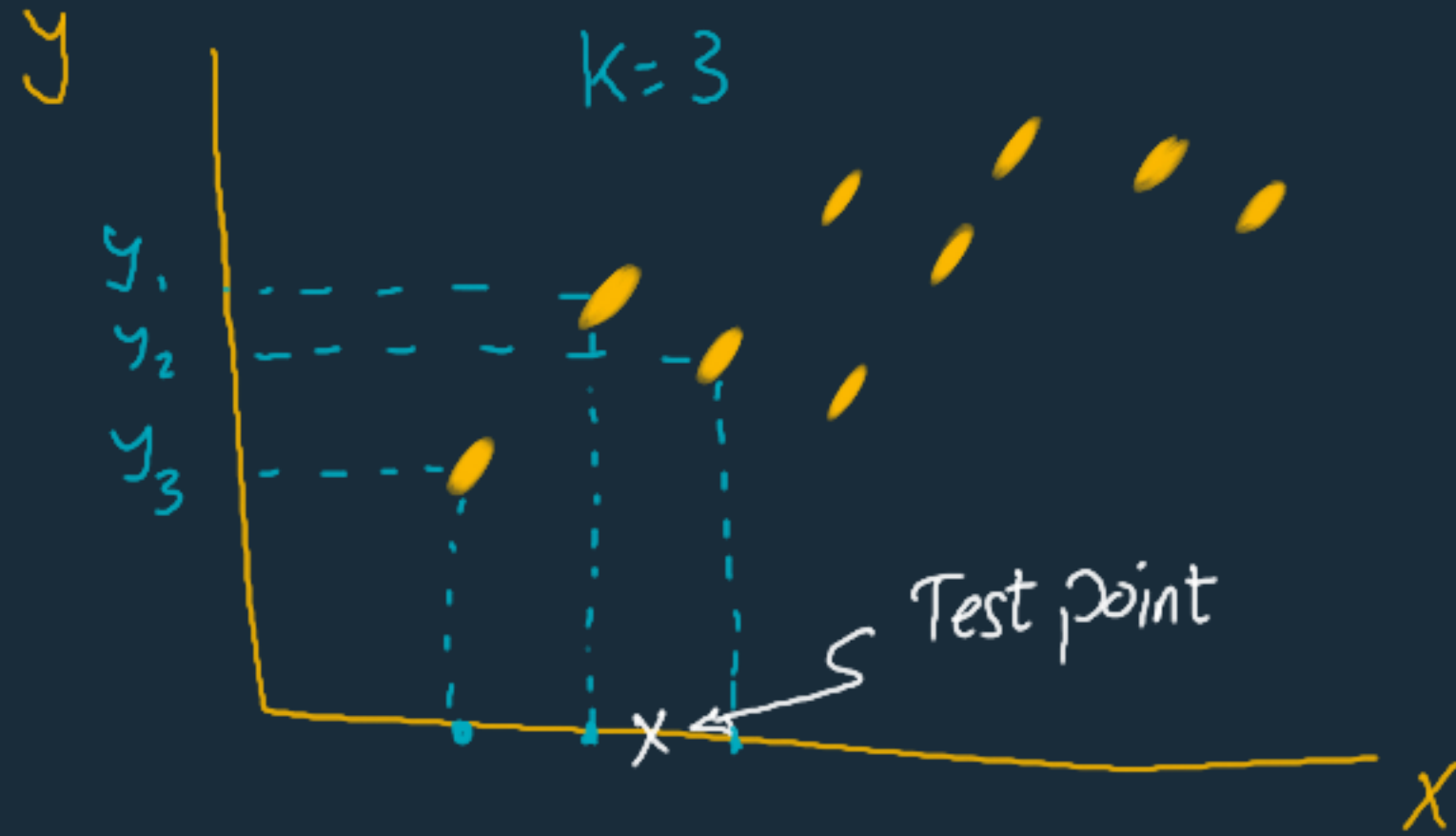


- Choose K
- Similarity
 - ↳ Nearest K
- Voting

Regressor



- choose K
- Similarity
 - ↳ Nearest K
- Average



$$\text{pred} = \frac{y_1 + y_2 + y_3}{3}$$

Knn Regressor

Training $\rightarrow$ Memorize Data

Prediction $\rightarrow$ Nearest K



Average

